

Analytics Programming Report

Analysing graduation pass rate with various factors across UK educational institutions for each of successive three years.

**Submitted by:
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1. Summary

- Brief summary in a few sentences of the major conclusions I have drawn.

This report incorporates the analysis of three different data files containing two different types of UK educational institutions for each of successive three years. This includes analysing the graduation pass rate with various factors which can be compiled here. Contact Learning Hours (CH) showed a weak correlation with Graduation Pass Rate but showed strong impact with Value Added Factor (VAF). Further Education Colleges (FE) vary more having high and low graduation rate compared to Sixth Form College (SF). SF outperformed FE constantly in both graduation rate and added factor. Analysis of regional and size variance highlights that institutions in South East and those categorized as Large has highest performance. Over three years, there were no changes only giving consistent results.

2. Methods

- Importing the data as well as identifying any anomalies and whether to include them in the analysis (max 1 page)

Sixth form (SF) csv file and further education csv file was imported using data steps including: input, label, format and further education (FE) tab file was imported using proc import. Further education csv and tab file were merged and sorted using ID as a unique key variable to combine related data into a single dataset. Both files were checked for any missing values using [proc means] and important missing value were replaced by mean or median method inside [proc stdize]. Some rows with missing critical values such as: Subtotal of each regions, Grand Value were removed using [where not] and [if not] because of its inconvenience and to ensure data quality so we can focus on valid observations. FE colleges had high number of outliers but were included after confirming they were not errors. Both sixth form and further education file was combined together in one table using [proc append]. But first I renamed every variable in both files to be same.

- Data summaries:

All the contents of the table were summarised to generate a metadata summary for appendix using [proc contents]. The metadata summary includes variables, their data types, length, and labels for their roles for transparency and reproducibility.

Some new variables were created for better analysis after data transformation and categorisation. CH per Learners for every year that represents the average contact hours were calculated to create a key variable for analysing institutional performance. It is used to compare institutions based on how much time learners spend on their studies can influence in Graduation pass rates. The results were rounded to one decimal place for simplicity. AvgCHperLearners, a new metric was calculated by taking mean of CHperLearners1, CHperLearners2, and CHperLearners3 that summarise the average contact learning hours across all three years. This average value simplifies the examination by providing a single value to signify the institution's overall contact hours.

To group institutions into size categories, new dataset CategCHperLearners was created which will allow us to analyse trends and differences in Graduation rates across created sizes. [Proc Sql] with case statement was used for data manipulation and categorization. In my code, I had encountered a problem where my Categorised Size was giving all the missing value. This happened because, before assigning any category I tried to run the code so later on this remained the same. I had to remove this Size column using [proc sql] drop command and used a new name Categorised_Size. To make my data more sorted, [proc sort] was used for the Categorised_Size and Institution Size was categorized as:

Average CH per Learners Range	Size Category
< 200	Small
>= 200 to < 400	Small-Medium
>= 400 to < 600	Medium
>= 600 to < 800	Large-Medium
>= 800	Large

- Exploratory data analysis

To understand the dataset's structure, distribution, and relationship between variables, exploratory data analysis was performed.

» Frequency Table:

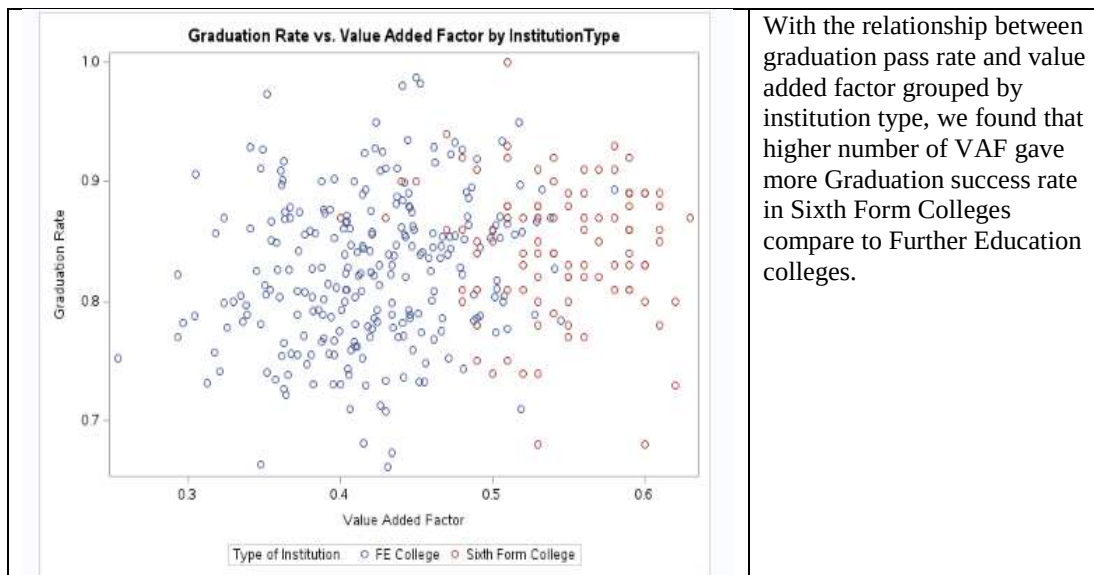
[Proc Freq] was used in exploratory data analysis to understand the distribution of key categorical variables. I had variables like: Region, Institution Type, Categorised Size with different values. Proc Freq helped us count how many times each different value appears in the dataset. For Institution type:

Institution Type	Frequency	Percent
FE College	254	71.75
Sixth Form College	100	28.25

» Box Plot:

[Proc Sgplot - vbox] was used to visualize the graphical distribution of Average CH per Learners, Graduation pass rate, Value Add Factor, Percent of Female Graduates across regions, institution size and type. I created vertical box plots grouped by Region, Institution Size, Institution Type. I checked distributions of all variables for outliers, skewness, missing data and relationship between key variables. It gave how CH per Learners, Graduation Rate, Value add Factor and Gender varies regionally.

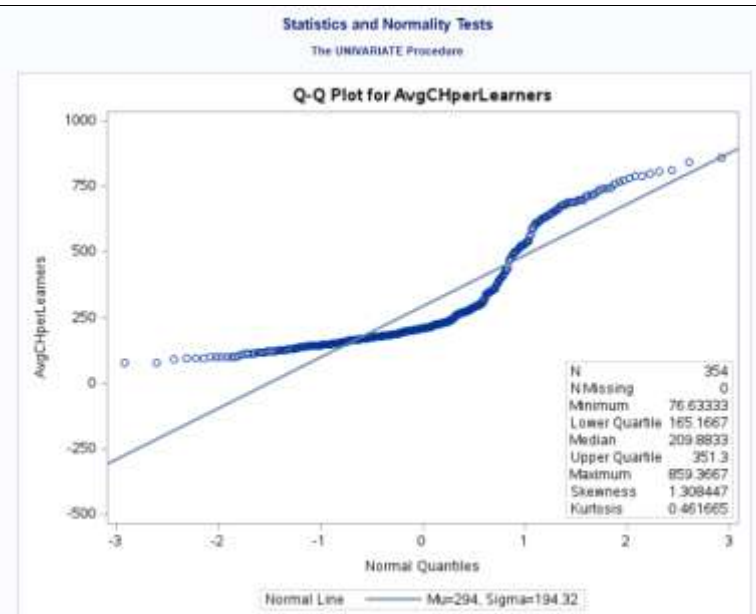
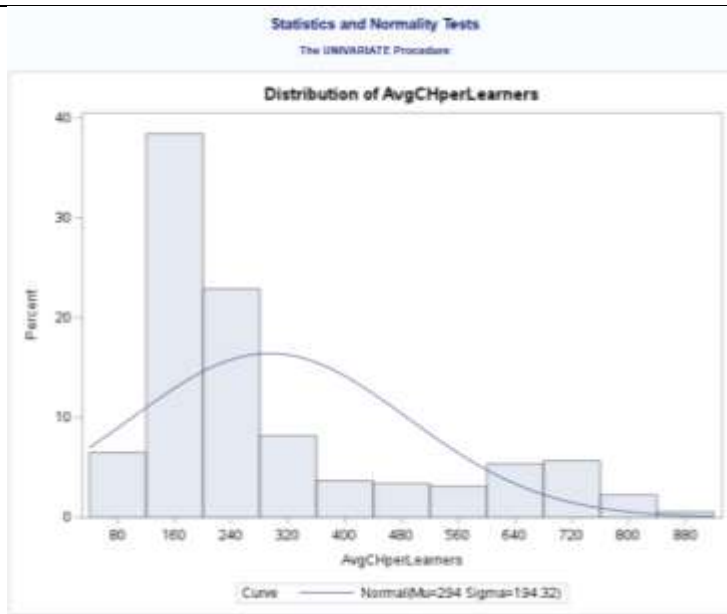
[Proc Sgplot – scatter] was used to see relations like: if colleges with more contact hours have higher graduation rates, is there link between value added factor and graduation success and if it differs by Institution Type?



» Normality Tests:

[Proc Univariate] was used to test if numerical variables: AvgCHperLearners, GradPR, VAF, GPercentFemale follow a normal distribution or not. Histogram, QQ-plots, Shapiro-Wilk tests were done to validate assumptions, helping to determine whether transformations was needed before parametric test or non-parametric methods are needed. In the result, we found that AvgCHperLearners was right skewed which indicates non-normality.

Variable	Skewness	Shapiro-Wilk (p-value)	Normality
GradPR	-0.1124	0.2656	Yes
VAF	0.182	0.0007	Yes
AvgCHperLearners	1.308	<0.0001	No



» Correlation Analysis:

[Proc Corr] was used to measure the strength and direction of relationship between numeric variables performing statistical analysis. Pearson and Spearman correlation coefficients were calculated for variables such as: AvgCHperLearners, GradPR, VAF, and GPercentFemale to identify significant connections. This helped us to find strong correlation between AvgCHperLearners and VAF so as to prioritize VAF to improve GradPR.

- Choice of statistical model and relevant validations.

As AvgCHperLearners was right-skewed, I did the data transformation using log but despite this. With only 4% variation with dependent and independent variable, the plot of residual and predicted not showing a random scatter suggested a poor fit of Linear Regression Model. When assumptions of normality and homogeneity of variance was violated, I decided to use non-parametric tests: Krusal-Wallis Test and Spearman Correlation Test.

» Krusal-Wallis Test:

We tested if independent variables (AvgCHperLearners, GradPR, VAF, GPercentFemale) differ by dependent variables (Region, institution type, categorized size, year) by comparing medians differences across groups.

[Proc npar1way] was used to generate box plot and statistics values which we can evaluate. We found a great significance of region on Average CH per Learners with p-value of 0.0012.

All variables mentioned above showed high significance with ($p < 0.0001$) between Further Education and Sixth Form Colleges where, sixth form constantly had high mean ranks.

While comparing with the categorized size, average CH per learners showed maximum difference with $\text{ChiSq} = 307.35$ and $p < 0.0001$. Categorised size didn't show much difference with Graduation pass rate ($p = 0.1609$).

When variables were analysed across three years, there seemed to be no difference at all like perfectly distributed group with $p\text{-value} = 1.00$.

Using [Proc Sgpanel] command, scatter plot was generated with GradPR on the X-axis with 3 different variables. Comparing CH per Learners with graduation rate across all 9 regions showed that 'South West' and 'North East' region had declined. But regions like: 'South East' and 'North West' has upward slope indicating if Contact Hours per learners are increased Graduation success rate might also increase in these regions, which show positive correlation.

Scatterplots of graduation pass rate and value added factor showed that the data points are spreaded out. So the correlation between these two variables are very low.

I had varied dependent variable whose plots will be attached in the results and interpretation.

I used [Proc Sgplot] to compare the median of different variables. Sixth Form College consistently show higher median GradPR of 75% in comparison with Further Education College. Small – Medium institutions display the largest box, indicating high variability in performance. Large institutions have smaller boxes showing more consistence GradPR. Some dots outside the boxes indicate the outliers in the dataset.

» Spearman Correlation Test:

As the dataset was not normally distributed, another non-parametric test that was done was to test the strength and direction of relationship between variables.

Using [Proc Corr] I found a strong positive relation with AvgCHperLearners and VAF. I used [Proc Sgscatter] to visually confirm correlations and detect outliers. Some key points that needs to be mentioned are:

AvgCHperLearners and VAF has upward sloping cloud with rho-value = 0.55 confirming our assumption of strong correlation. Like I predicted, GradPR and VAF showed very weak drift aligning with only 0.22 as p-value. From the histogram of GPercentFemale, we can finalise that female graduates are tightly gathered around 50%.

I also used [Proc Freq] to check how results has changed over time. For both sixth form and further education, both graduation rate and contact hour per learners has p-value = 1.00 with no distinctive change.

3. Results and interpretation (max 1 page for each statistical method used)

» Krusal-Wallis Test:

By Institution Type:

- i. All variables showed notable differences with ($p < 0.0001$).
- ii. Sixth Form has higher ranks than FE colleges.
- iii. Institution Type has great effect on AvgCHperLearners with ChiSq = 190.01.

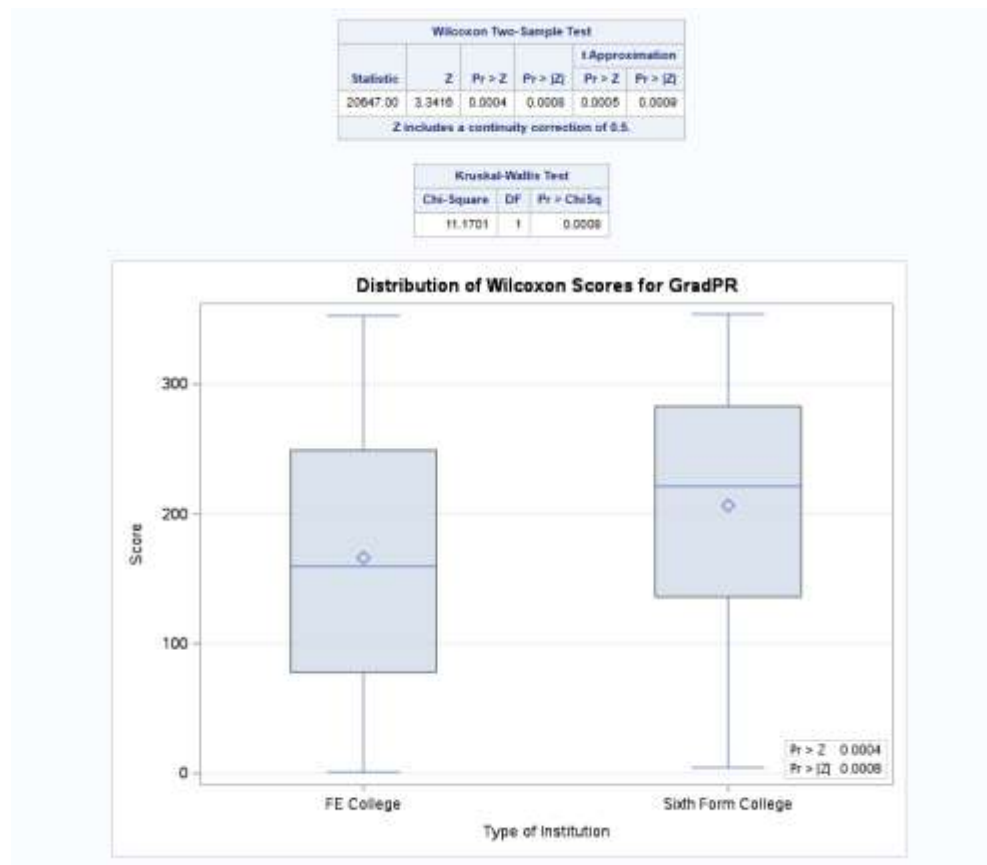
By Region:

- i. Region showed most significance to AvgCHperLearners with p-value of 0.0012.
- ii. It wasn't any relationship with female graduates.

By Size:

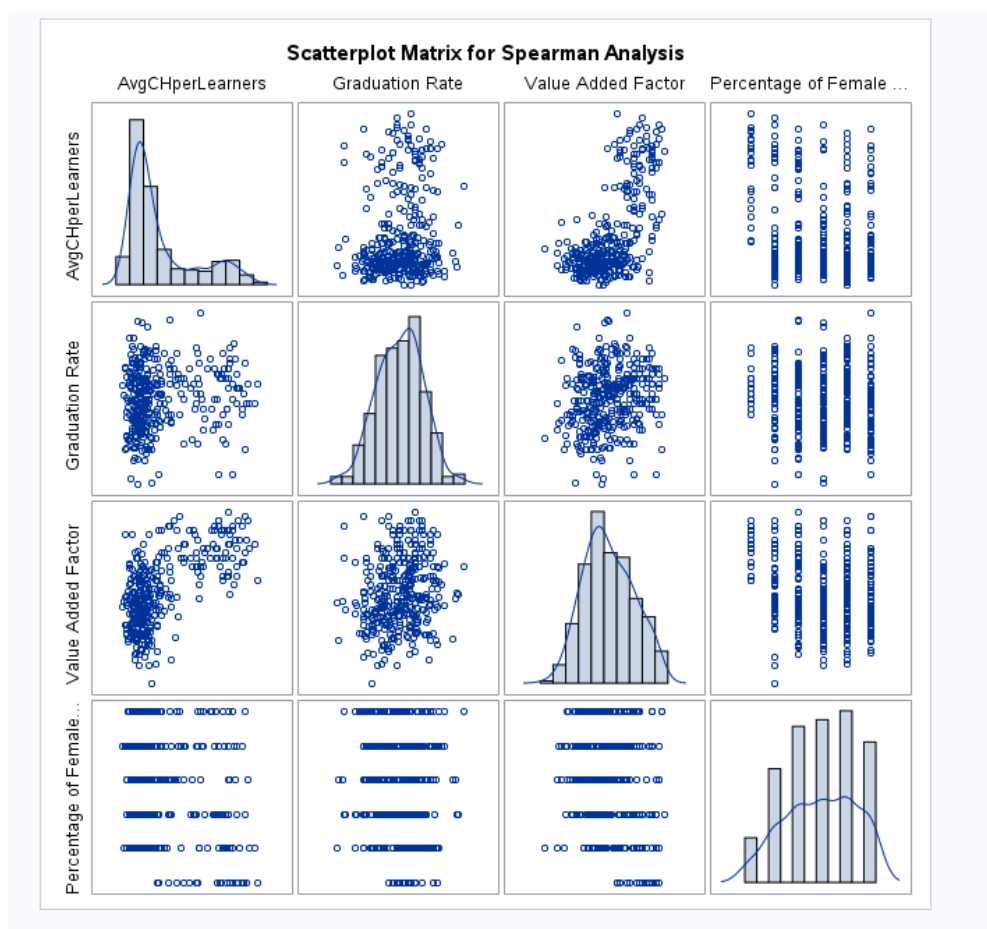
- i. Categorised size was closely related to CH per Learners.

Other than region effecting Average CH per Learners, Institution Type has the most notable difference in the performance.



» Spearman Correlation Test:

- i. AvgCHperLearners and VAF: $\rho = 0.55$ ($p < 0.0001$)
With extra added help, learners show better outcome giving high contact hours.
- ii. AvgCHperLearners and GradPR: $\rho = 0.13$
Its shows weak relationship and does not impact graduation rate.
- iii. VAF and GPercentFemale: $\rho = 0.16$ ($p = 0.0002$)
Its shows that with higher female graduating, added value seems to decline.
- iv. GradPR, AvgCHperLearners and Year: $\rho = 1.00$
This shows that both rates remained constant each year for both FE and SixthForm Colleges.
- v. GradPR : Sixth Form College did better performance in compare to FE college with median positive 15% for Large Size Institution.
- vi. VAF: Large Institutions displayed lowest variance of 5%. Compare to Institution size, institution type has more weightage.



Validation:

Test	Assumptions	Met or not?	Evidence
Kruskal-Wallis	Non-normal, independent data	Yes	Histogram, boxplots
Spearman Correlation	Monotonic relationships	Yes	Scatterplot matrix

4. Conclusions

Insights we got from interpretation of our results, the conclusion can be drawn in many points. As contact learning hours directly related to additional supports, Further Education Colleges needs to target on those factors to excel in the curriculum. Rather than focus on increasing learning hours, colleges should focus on other factors like better mentorship and curriculum relevance. Large institution in regions like South East showed good performance which might be due to easy access of resources. If concerned people can make sure of distributing the fund equally, all areas will have a good graduation rate in no time.

5. Appendix

Metadata Summary

The CONTENTS Procedure

Data Set Name	MYLIB.CATEGCHPERLEARNERS	Observations	354
Member Type	DATA	Variables	17
Engine	V9	Indexes	0
Created	25/03/2025 08:46:29	Observation Length	232
Last Modified	25/03/2025 08:46:29	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	YES
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	564
Obs in First Data Page	354
Number of Data Set Repairs	0
Filename	
Release Created	9.0401M7
Host Created	Linux
Inode Number	2731582948
Access Permission	rw-r--r--
Owner Name	
File Size	256KB
File Size (bytes)	262144

Missing Values.

The MEANS Procedure

Variable	Label	N	N Miss	Minimum	Maximum	Mean	Std Dev
Total_CH_Year1	Total CH Year 1	354	0	150890.00	5877756.00	1738689.88	975899.00
Leamers1	Leamers Year 1	354	0	426.0000000	44406.00	9431.31	7395.31
Total_CH_Year2	Total CH Year 2	354	0	151298.00	7716710.00	1772777.60	1041608.88
Leamers2	Leamers Year 2	354	0	453.0000000	50723.00	8114.18	6522.74
Total_CH_Year3	Total CH Year 3	354	0	164005.00	7545209.00	1652436.22	980363.99
Leamers3	Leamers Year 3	354	0	583.0000000	53281.00	7851.00	6394.43
GradPR	Graduation Rate	354	0	0.8613971	1.0000000	0.8305394	0.0608576
VAf	Value Added Factor	354	0	0.2540364	0.6300000	0.4552440	0.0770672
GPercentFemale	Percentage of Female Graduates	354	0	47.0000000	52.0000000	49.9180791	1.4870778
GPercentMale	Percentage of Male Graduates	354	0	48.0000000	53.0000000	50.0819209	1.4870778
CHperLeamers1		354	0	60.6000000	937.3000000	285.7528249	212.0705488
CHperLeamers2		354	0	73.0000000	875.5000000	305.5149718	193.9312431
CHperLeamers3		354	0	70.6000000	830.9000000	290.7435028	183.2566646
AvgCHperLeamers		354	0	76.6333333	859.3666667	294.0037665	194.3177122

Variables in Creation Order					
#	Variable	Type	Len	Format	Label
1	Total_CH_Year1	Num	8	COMMA12.	Total CH Year 1
2	Learners1	Num	8	COMMA12.	Learners Year 1
3	Total_CH_Year2	Num	8	COMMA12.	Total CH Year 2
4	Learners2	Num	8	COMMA12.	Learners Year 2
5	Total_CH_Year3	Num	8	COMMA12.	Total CH Year 3
6	Learners3	Num	8	COMMA12.	Learners Year 3
7	GradPR	Num	8	8.2	Graduation Rate
8	VAF	Num	8	8.2	Value Added Factor
9	InstitutionType	Char	50		Type of Institution
10	Region	Char	50		Geographical Region
11	GPercentFemale	Num	8	8.2	Percentage of Female Graduates
12	GPercentMale	Num	8	8.2	Percentage of Male Graduates
13	CHperLearners1	Num	8		
14	CHperLearners2	Num	8		
15	CHperLearners3	Num	8		
16	AvgCHperLearners	Num	8		
17	Categorised_Size	Char	15		

Sort Information	
Sortedby	InstitutionType Region Categorised_Size
Validated	YES
Character Set	ASCII

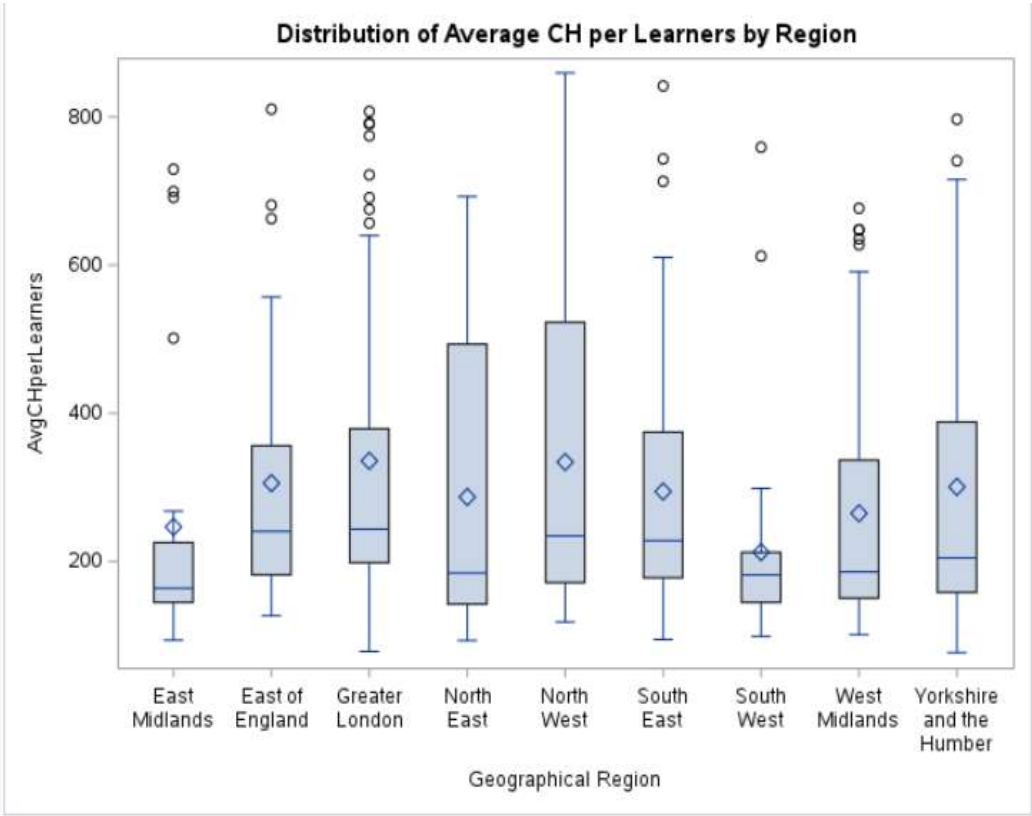
Categorical Variable Distribution

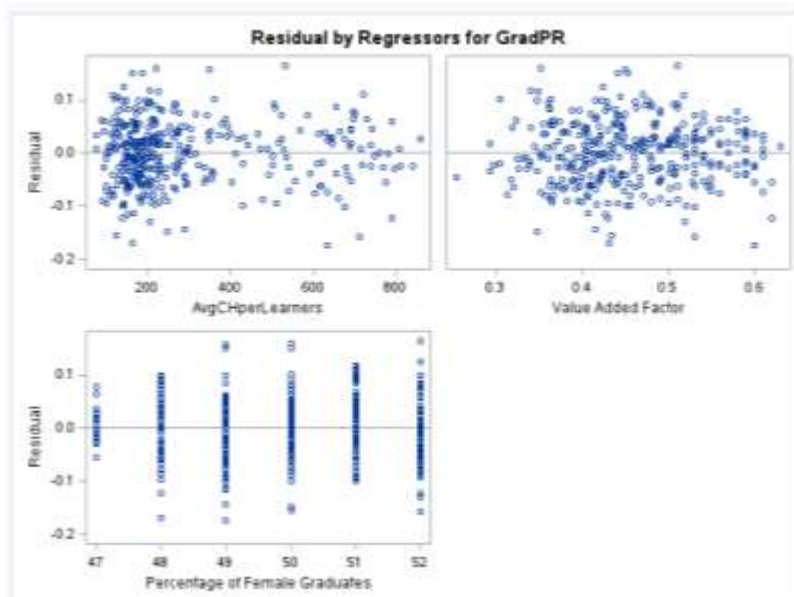
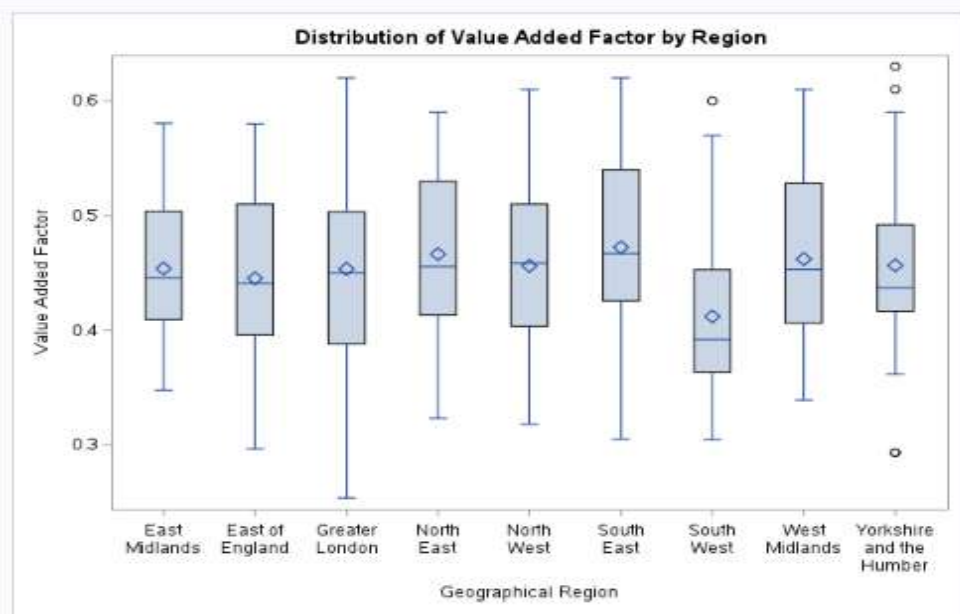
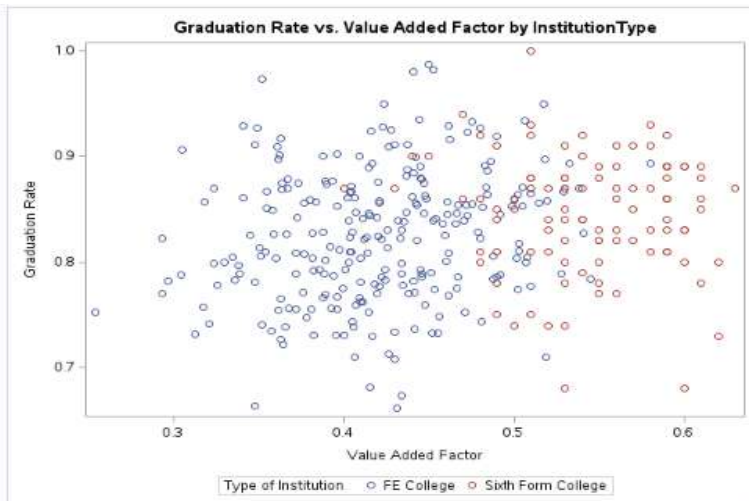
The FREQ Procedure

Geographical Region		
Region	Frequency	Percent
East Midlands	24	6.78
East of England	31	8.76
Greater London	47	13.28
North East	21	5.93
North West	61	17.23
South East	59	16.67
South West	29	8.19
West Midlands	45	12.71
Yorkshire and the Humber	37	10.45

Type of Institution		
InstitutionType	Frequency	Percent
FE College	254	71.75
Sixth Form College	100	28.25

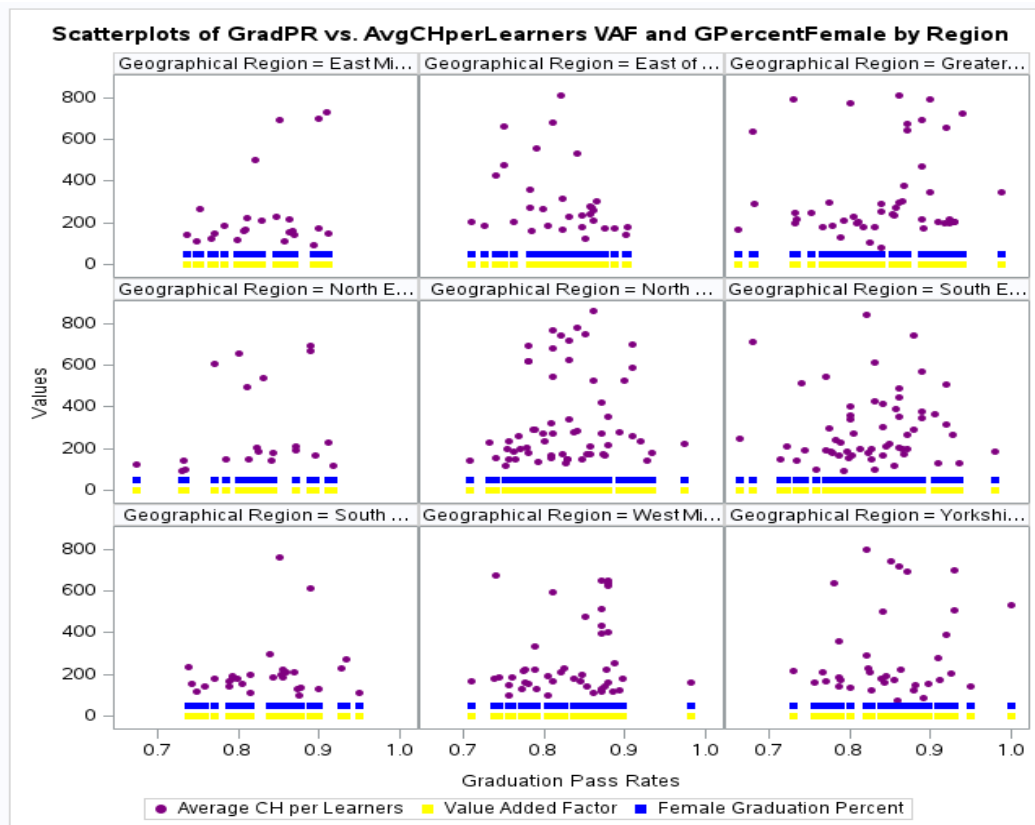
Categorised_Size	Frequency	Percent
Large	4	1.13
Large-Medium	45	12.71
Medium	29	8.19
Small	159	44.92
Small-Medium	117	33.05

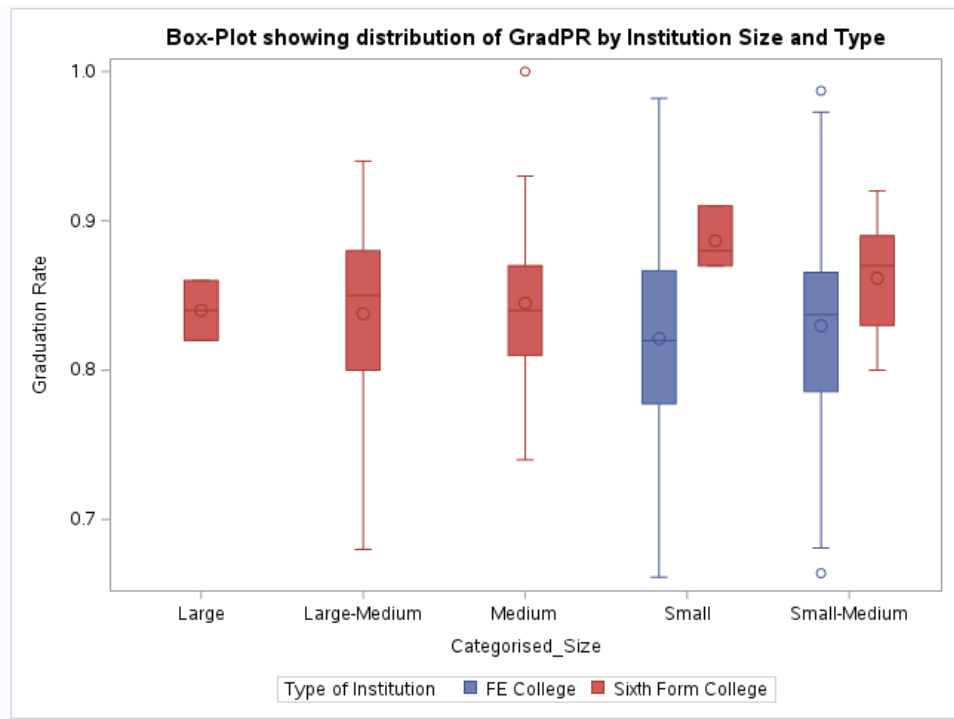
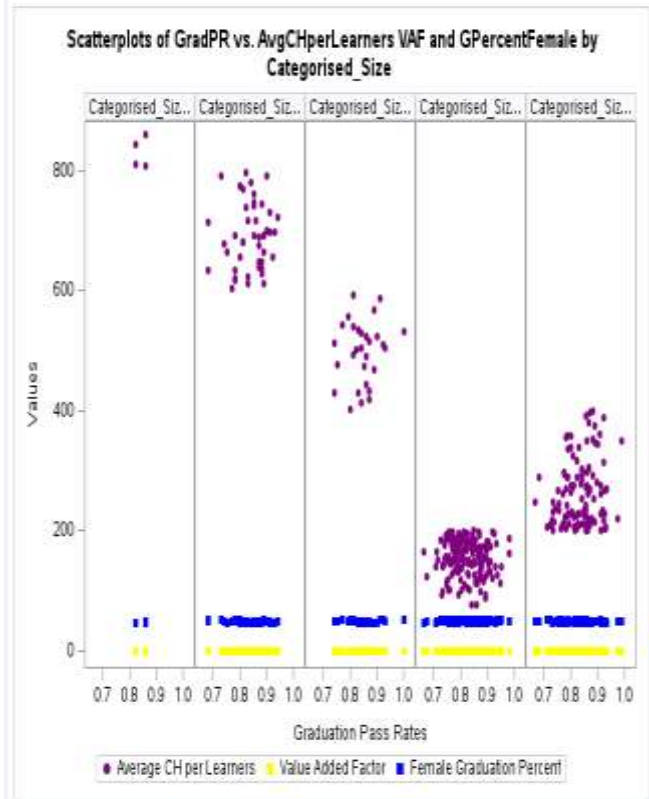
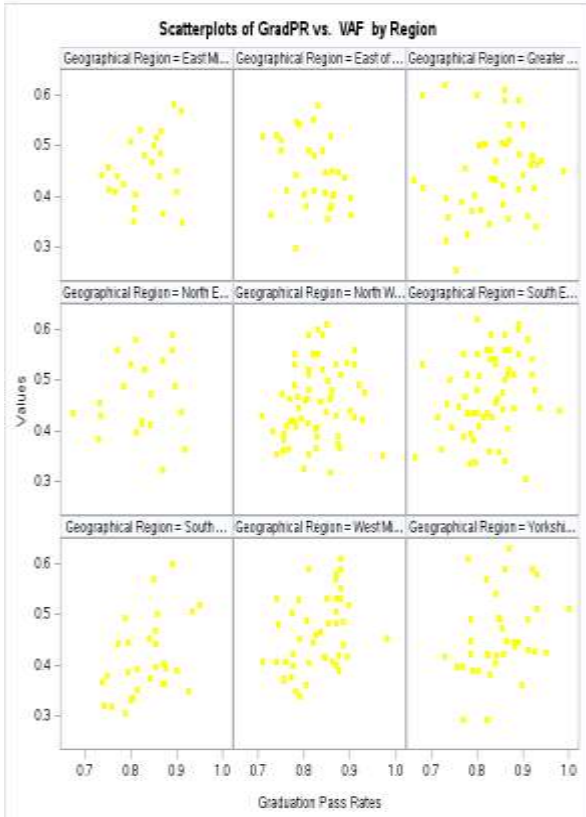




Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
InstitutionType	1	1041	71.61	<.0001
Region	8	1041	1.63	0.1107
Categorised_Size	4	1041	836.41	<.0001
year	2	1041	14.20	<.0001
GPercentFemale	5	1041	3.66	0.0027

Least Squares Means										
Effect	Type of Institution	Geographical Region	Categorised_Size	year	Percentage of Female Graduates	Estimate	Standard Error	DF	t Value	Pr > t
InstitutionType	FE College					446.31	5.7126	1041	78.13	<.0001
InstitutionType	Sixth Form College					507.10	4.3807	1041	115.76	<.0001
Region		East Midlands				470.56	6.0601	1041	67.42	<.0001
Region		East of England				482.52	6.0554	1041	79.68	<.0001
Region		Greater London				485.11	5.2854	1041	91.78	<.0001
Region		North East				463.20	7.3154	1041	63.32	<.0001
Region		North West				483.71	4.9185	1041	98.35	<.0001
Region		South East				479.45	5.2029	1041	92.15	<.0001
Region		South West				472.52	6.5429	1041	72.22	<.0001
Region		West Midlands				474.52	5.8194	1041	84.44	<.0001
Region		Yorkshire and the Humber				478.77	5.8066	1041	81.20	<.0001
Categorised_Size			Large			792.47	15.4246	1041	51.38	<.0001
Categorised_Size			Large-Medium			658.91	5.7360	1041	114.87	<.0001
Categorised_Size			Medium			466.89	6.5834	1041	70.92	<.0001
Categorised_Size			Small			185.19	4.2994	1041	43.07	<.0001
Categorised_Size			Small-Medium			280.08	3.8797	1041	72.19	<.0001
year				1		468.46	4.2390	1041	110.51	<.0001
year				2		488.22	4.2390	1041	115.17	<.0001
year				3		473.45	4.2390	1041	111.89	<.0001
GPercentFemale					47.00	485.62	7.7266	1041	62.85	<.0001
GPercentFemale					48.00	470.55	5.0370	1041	93.42	<.0001
GPercentFemale					49.00	471.44	4.8914	1041	96.38	<.0001
GPercentFemale					50.00	483.61	4.9220	1041	98.25	<.0001
GPercentFemale					51.00	467.41	4.8014	1041	97.35	<.0001
GPercentFemale					52.00	481.62	5.1163	1041	94.13	<.0001





Scatterplot Matrix for Spearman Analysis

The FREQ Procedure

Type of Institution=FE College
Summary Statistics for GradPR by year

Cochran-Mantel-Haenszel Statistics (Based on Rank Scores)				
Statistic	Alternative Hypothesis	DF	Value	Prob
1	Nonzero Correlation	1	0.0000	1.0000
2	Row Mean Scores Differ	29	0.0000	1.0000

Total Sample Size = 762

Scatterplot Matrix for Spearman Analysis

The FREQ Procedure

Type of Institution=Sixth Form College
Summary Statistics for GradPR by year

Cochran-Mantel-Haenszel Statistics (Based on Rank Scores)				
Statistic	Alternative Hypothesis	DF	Value	Prob
1	Nonzero Correlation	1	0.0000	1.0000
2	Row Mean Scores Differ	22	0.0000	1.0000

Total Sample Size = 300

Scatterplot Matrix for Spearman Analysis

The FREQ Procedure

Type of Institution=FE College
Summary Statistics for AvgCHperLearners by year

Cochran-Mantel-Haenszel Statistics (Based on Rank Scores)				
Statistic	Alternative Hypothesis	DF	Value	Prob
1	Nonzero Correlation	1	0.0000	1.0000
2	Row Mean Scores Differ	250	0.0000	1.0000

Total Sample Size = 762

Spearman Rank Correlations

The CORR Procedure

4 Variables: AvgCHperLearners GradPR VAF GPercentFemale

Simple Statistics							
Variable	N	Mean	Std Dev	Median	Minimum	Maximum	Label
AvgCHperLearners	354	294.00377	164.31771	209.86333	76.63333	859.38667	
GradPR	354	0.83054	0.06086	0.83066	0.66140	1.00000	Graduation Rate
VAF	354	0.46524	0.07707	0.44660	0.25404	0.83000	Value Added Factor
GPercentFemale	354	48.81808	1.46708	50.00000	47.00000	52.00000	Percentage of Female Graduates

Spearman Correlation Coefficients, N = 354				
Prob > r under H0: Rho=0				
	AvgCHperLearners	GradPR	VAF	GPercentFemale
AvgCHperLearners	1.00000	0.13091	0.55373	-0.22867
		0.0137	<.0001	<.0001
GradPR	0.13091	1.00000	0.22245	-0.07477
Graduation Rate	0.0137		<.0001	0.1604
VAF	0.55373	0.22245	1.00000	-0.19406
Value Added Factor	<.0001	<.0001		0.0002
GPercentFemale	-0.22867	-0.07477	-0.19406	1.00000
Percentage of Female Graduates	<.0001	0.1604	0.0002	